

CSS326 Project Proposal

Project name

GrocerEase

Group Members

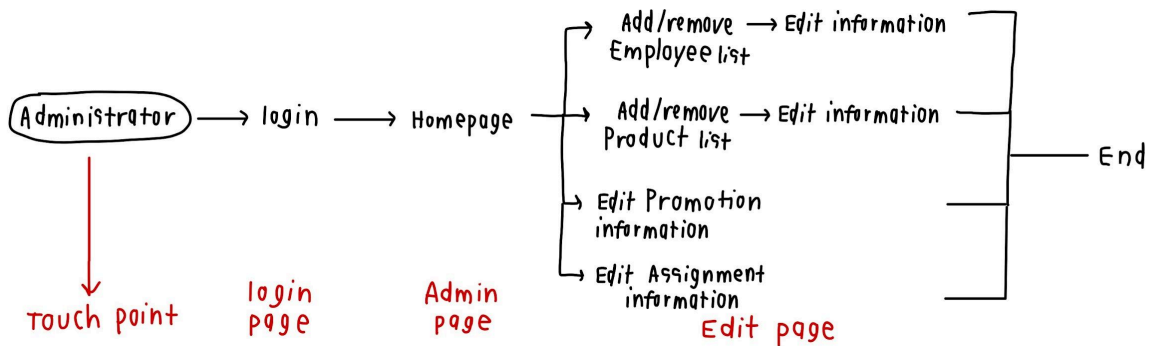
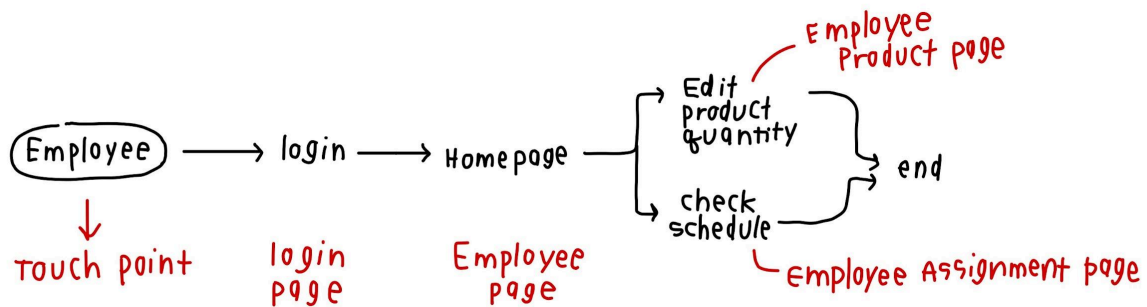
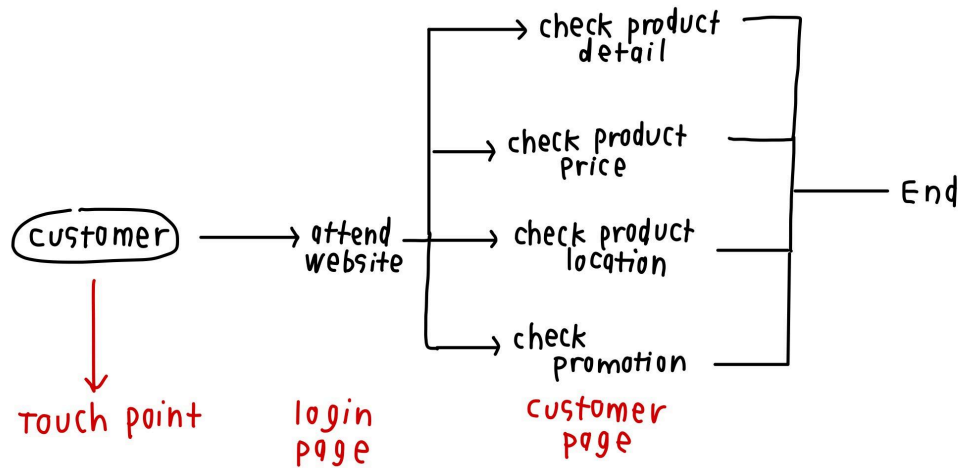
Student Name – Student ID – Contribution/Roles

1. Ms. Kawisara Premwinai – 6522770708
2. Ms. Palinee Sadibhakdi – 6522770724
3. Mr. Jakrapat Wongpradu – 6522771045

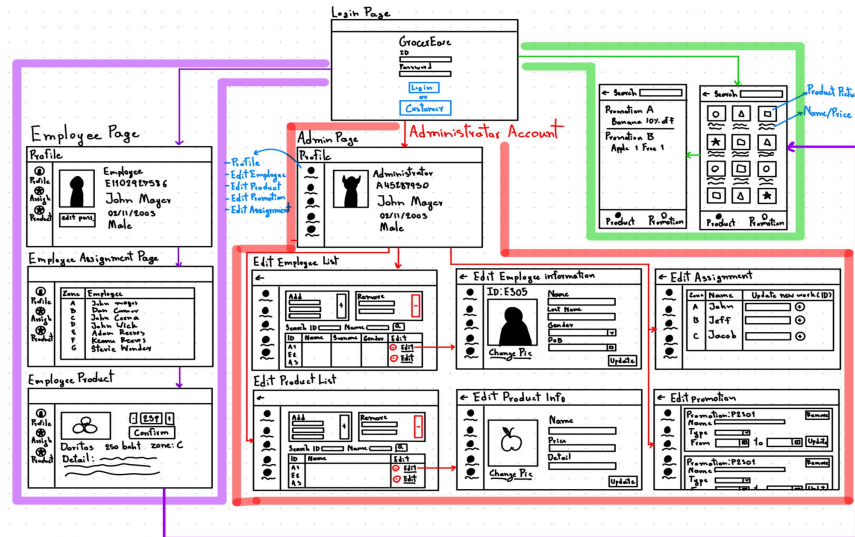
Project Description

Our project is the **Grocery Management System**. It aimed at improving operational efficiency within a Grocery store environment. The system is tailored for employees, customers, and administrators. For employees and administrators, it enables them to manage various tasks ,which are about the employees schedule and product stocking, for the smooth running of the store. For customers, it enables them to easily find the products they want and they can also check about the promotion in the supermarket. We used MySQL as the database to securely store and retrieve information while ensuring smooth performance. The front-end was developed using HTML and CSS to create an interface that is simple, responsive, and easy to use. Additionally, SQL procedures were implemented to handle data efficiently and improve the system's functionality.

User Journey



System Structure



Customer Functions

- Browse the website as a guest
- Find a product
- Check product information
- Look up promotions

Employee Functions

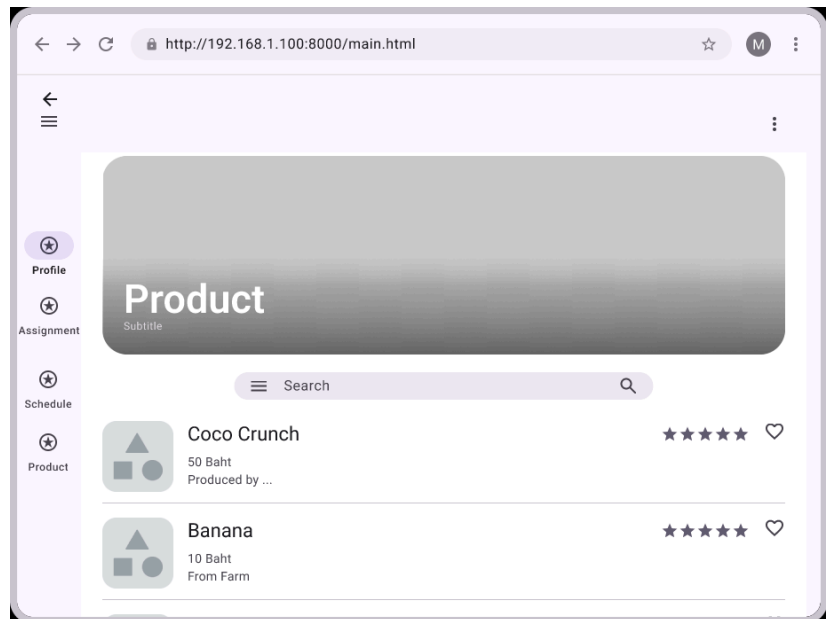
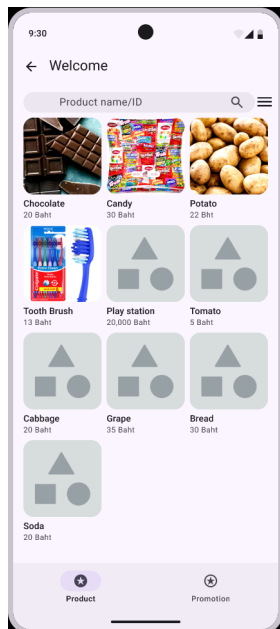
- Login to employee account
- View employee profile
- Find a product
- Check product information
- Edit quantity
- View assigned location
- Change password

Admin Functions

- Login to admin account
- Admin profile
- Add/Delete Account
- Edit account information
- Add/Delete product
- Edit product information
- Assigned employee to location
- Add/Delete Promotion
- Edit promotion information
- Change password

Initial User Interface

A web browser window displaying a login page. The address bar shows the URL `http://192.168.1.100:8000/login.html`. The page has a light purple background. At the top center is the title "Login". Below it are two input fields: "Username" and "Password", each with a small circular icon containing an 'x' on the right. Below the password field is a "Login" button. Underneath the button is a text input field with a placeholder "Qr" and a dashed line. At the bottom is a "Customer" button.

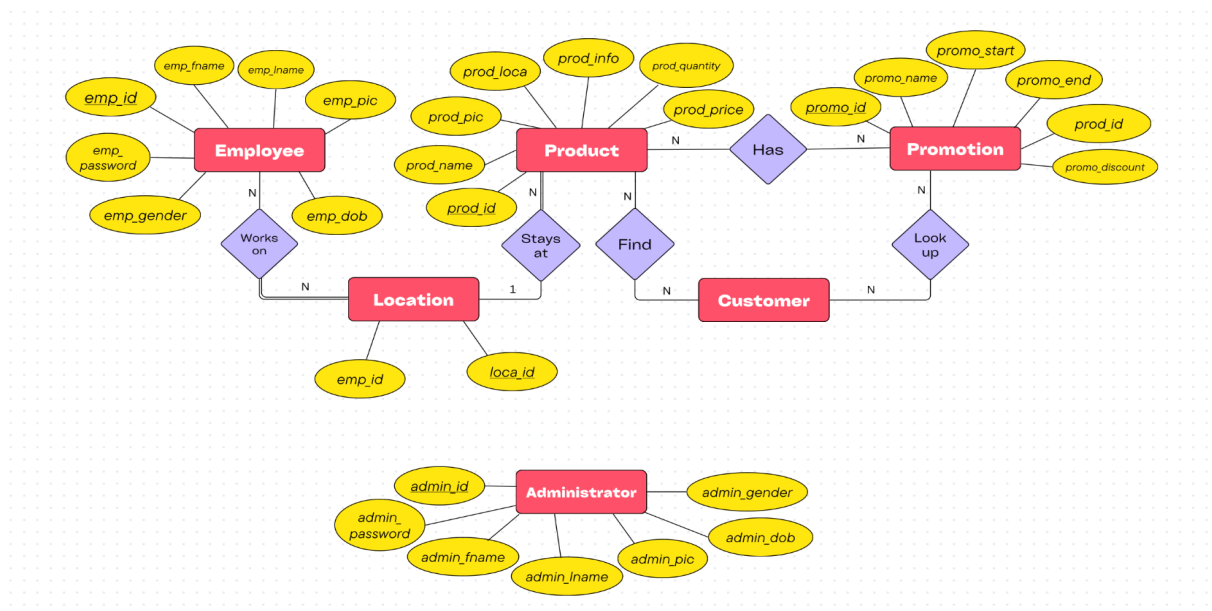


Database Design

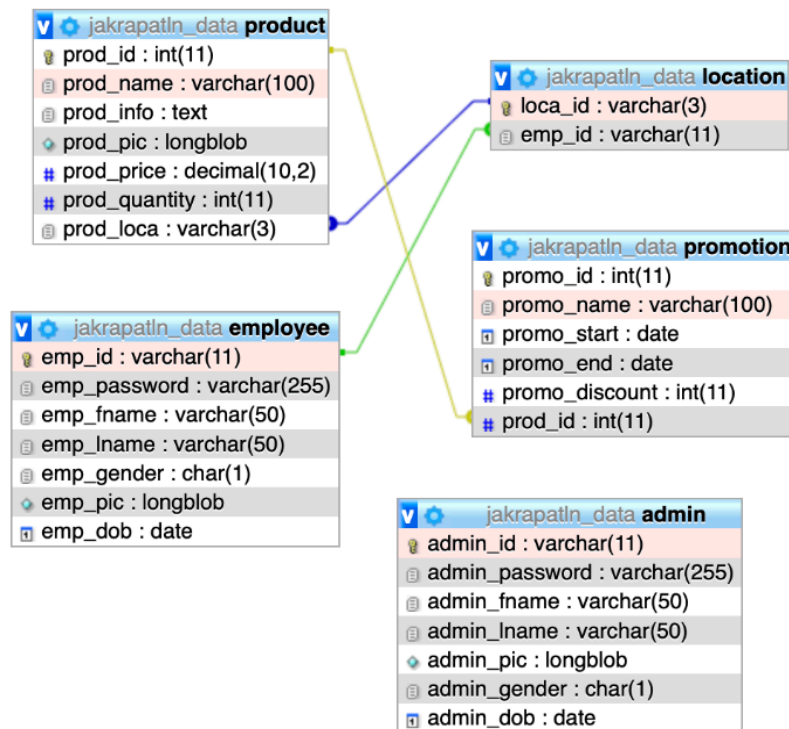
Entities list

- Employee
- Product
- Promotion
- Location
- Customer
- Administrator

ER Diagram



Physical Design for MySQL



What are changes from the initial proposal?

Implementation of Database Management System

The database management system (DBMS) implemented for developing the GrocerEase system is MySQL. Other tools for developing the application are HTML, CSS, and PHP. Source code of this project is uploaded to GitHub and can be accessed in this link:

<https://github.com/danikpw/GrocerEase.git>

Website link: <http://jakrapat.lnw.mn/login.php>

Data Tables

Admin table in GrocerEase database

				admin_id	admin_password	admin_fname	admin_lname	admin_pic	admin_gender	admin_dob
☐				A1	f4f830af345ba0f928239d2eed08c2173ceb2a4fb7b0deecc8...	Jakrapat	Wongpradu	[BLOB - 1.9 MiB]	M	2003-11-02
☐				A2	b9c950640e1b3740e98acb93e669c65766f6670dd1609ba91f...	Kawisara	Premwinai	[BLOB - 2.4 MiB]	F	2004-07-15
☐				A3	b9c950640e1b3740e98acb93e669c65766f6670dd1609ba91f...	Palinee	Smith	[BLOB - 1.6 MiB]	F	2024-11-07

Employee table in GrocerEase database

				emp_id	emp_password	emp_fname	emp_lname	emp_gender	emp_pic	emp_dob
☐				E1	f4f830af345ba0f928239d2eed08c2173ceb2a4fb7b0deecc8...	Jakrapat	Wongpradu	M	[BLOB - 267.0 KiB]	2024-11-15
☐				E2	b9c950640e1b3740e98acb93e669c65766f6670dd1609ba91f...	Mike	Ross	M	[BLOB - 332.6 KiB]	2024-11-05
☐				E4	b9c950640e1b3740e98acb93e669c65766f6670dd1609ba91f...	Alicia	Keys	F	[BLOB - 176.1 KiB]	2024-11-10
☐				E5	b9c950640e1b3740e98acb93e669c65766f6670dd1609ba91f...	Kawisara	Premwinai	F	[BLOB - 44.5 KiB]	2024-11-12

Location table in GrocerEase database

				loca_id	emp_id
☐				D	E1
☐				A	E2
☐				B	E2
☐				E	E2
☐				G	E4
☐				C	E5
☐				F	E5

Product table in GrocerEase database

				prod_id	prod_name	prod_info	prod_pic	prod_price	prod_quantity	prod_loca
☐				1	Joyride	Sweet jelly beans	[BLOB - 171.9 KiB]	12.50	312	A
☐				2	Grape	Green grapes sour	[BLOB - 66.0 KiB]	175.00	888	C
☐				3	Tomato	Fresh Red Tomatoes	[BLOB - 30.9 KiB]	16.00	1600	C
☐				4	Potato	Potato, Smells good.	[BLOB - 117.7 KiB]	37.75	221	E
☐				5	Porsche	Porsche 911 GT3RS	[BLOB - 1.0 MiB]	19000000.00	2	E
☐				6	ToothBrush	Toothbrush 3 pieces	[BLOB - 272.9 KiB]	14.75	236	A
☐				7	Hersheys	Chocolate bar	[BLOB - 69.4 KiB]	39.99	450	B
☐				8	BankCrocs	Recharge-able power bank.	[BLOB - 131.9 KiB]	1200.00	36	C

Stored Procedures

In the GrocerEase database, we included 2 stored procedures to help our project database. These 2 stored procedures are shown below.

1. “GetLocationsWithEmployees” procedure

In this procedure, it retrieves a list of locations along with the employees assigned to each location. It includes details such as the employee's first name, last name, gender, picture, and date of birth. The procedure ensures all locations are listed, even if no employee is currently assigned to them, by using a LEFT JOIN.

```
BEGIN
    SELECT l.loc_a_id, l.emp_id AS LocationEmpID, e.emp_fname, e.emp_lname,
    e.emp_gender, e.emp_pic, e.emp_dob
    FROM location l
    LEFT JOIN employee e ON l.emp_id = e.emp_id
    ORDER BY l.loc_a_id ASC;
END
```

Routine name	GetLocationsWithEmploy					
Type	PROCEDURE					
Parameters	<table><thead><tr><th>Direction</th><th>Name</th><th>Type</th><th>Length/Values</th><th>Options</th></tr></thead></table>	Direction	Name	Type	Length/Values	Options
Direction	Name	Type	Length/Values	Options		
Definition	Add parameter <pre>1 BEGIN 2 SELECT 3 l.loc_a_id, 4 l.emp_id AS LocationEmpID, 5 e.emp_fname, 6 e.emp_lname, 7 e.emp_gender, 8 e.emp_pic, 9 e.emp_dob 10 FROM 11 location l 12 LEFT JOIN 13 employee e 14 ON 15 l.emp_id = e.emp_id 16 ORDER BY 17 l.loc_a_id ASC; 18 END</pre>					

2. “ GetPromotionProductInfo “ procedure

In this procedure, it retrieves information about promotions along with the associated product details. It includes data such as the promotion name, start and end dates, discount percentage, and the product's name, price, picture, and description. The procedure ensures all promotions are listed, even if a promotion is not linked to a specific product, by using a LEFT JOIN.

BEGIN

```
SELECT d.promo_id, d.promo_name, d.promo_start, d.promo_end,  
d.promo_discount, d.prod_id AS promoprod_id, p.prod_name, p.prod_price, p.prod_pic,  
p.prod_info
```

```
FROM promotion d
```

```
LEFT JOIN product p ON d.prod_id = p.prod_id
```

```
ORDER BY d.promo_id ASC;
```

END

Type	PROCEDURE					
Parameters	<table><thead><tr><th>Direction</th><th>Name</th><th>Type</th><th>Length/Values</th><th>Options</th></tr></thead></table>	Direction	Name	Type	Length/Values	Options
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The problem that we found during development ?

- Incorrect diagram design affects the system design.
- Multiple rounds of interface revision
- User Experience (UX) Flaws
- HTML is not rendered, so we need to clear the cache.
- Database Connection Problems
- Deployment Issues

Team Contributions

1.Kawisara Premwinai 6522770708

- Front-end developer: Construct the website and manage the data from the back-end to front end.

2.Palinee Sadibhakdi 6522770724

- Database developer: Design and construct the database, Create stored procedures for querying the data, writing SQL query commands.

- UX/UI Designer: Design website interface

3.Jakrapat Wongpradu 6522771045

- Back-end developer: Construct connections between website and database.